ZIP Code-Aware Sale Price Prediction

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DSC680-T301 Applied Data Science • Week 4, Milestone 3

A screening estimate for homes that already sold

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Three questions

A national median prices the bathroom the same in every ZIP.

- How close can beds, baths, size, lot, and ZIP get to the sale price?
- How much does ZIP help, versus no location and versus the ZIP median?
- Does the miss change by market, and by how many sales that ZIP has?

The cost of being wrong: too high stalls the deal. Too low leaves money on the table.

Who the 15% describes

- 812,009 sold listings. About 25% dropped for missing beds, baths, size, or lot.
- Dropped homes are cheaper: median $249,000 versus $370,000 kept.
- Florida is 17% of the dropped rows and 9% of the kept rows. California moves the other way.
- ZIPs with fewer than 20 sales: median $250,000 versus $380,000 in the ZIPs that are scored.

Headline error is for documented houses in deeper, higher-priced ZIPs.

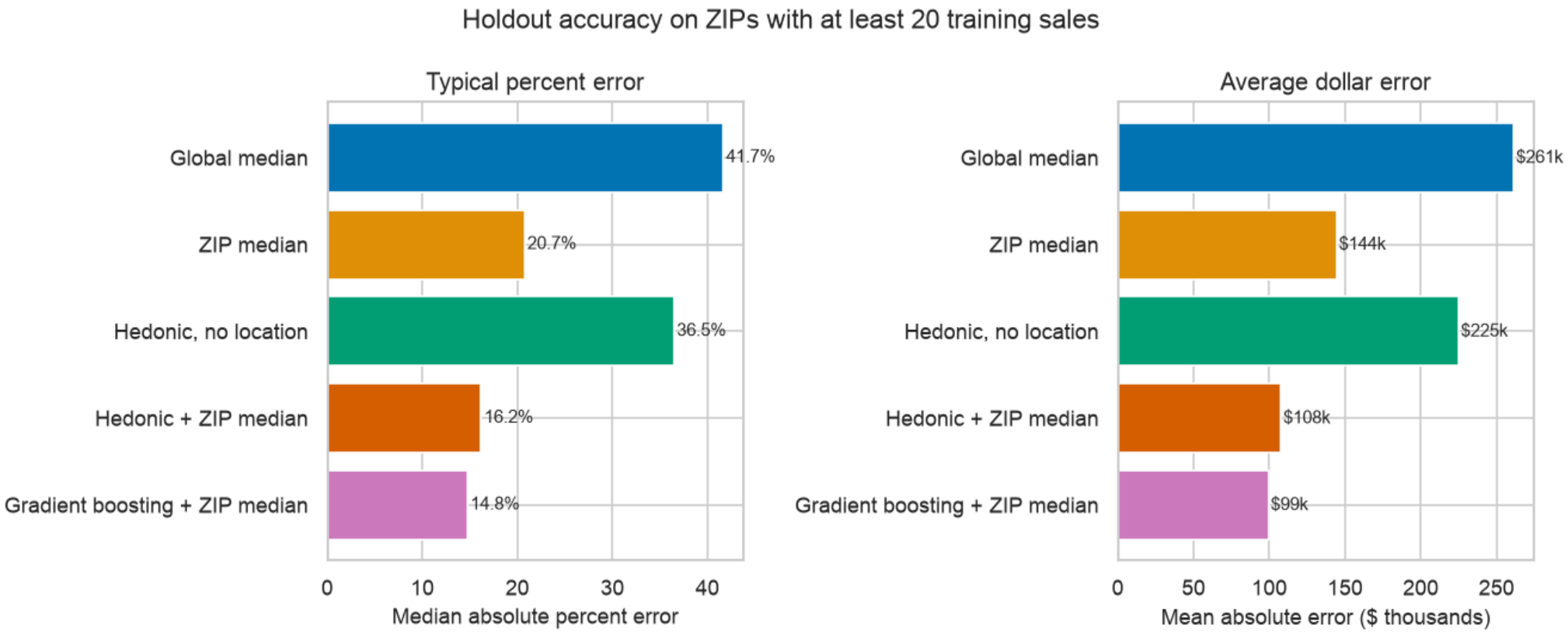
What was compared

Fit on log price. Score in dollars. Headline metric: typical percent miss.

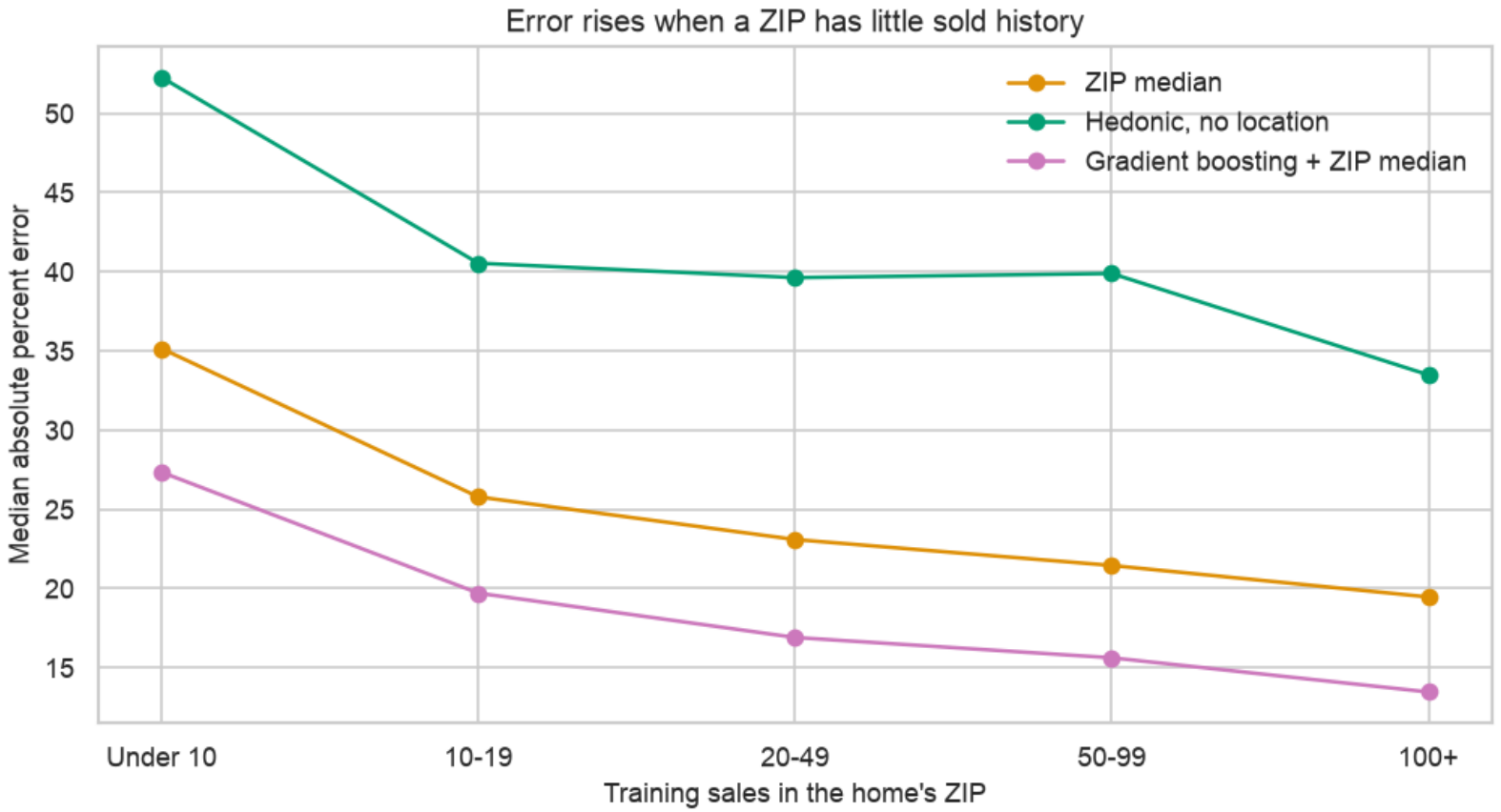
- National median
- ZIP median, from training sales only
- Regression on beds, baths, size, and lot
- That regression, plus the ZIP median
- Gradient boosting on the same inputs

Five-fold check rebuilds the ZIP median inside each fold.

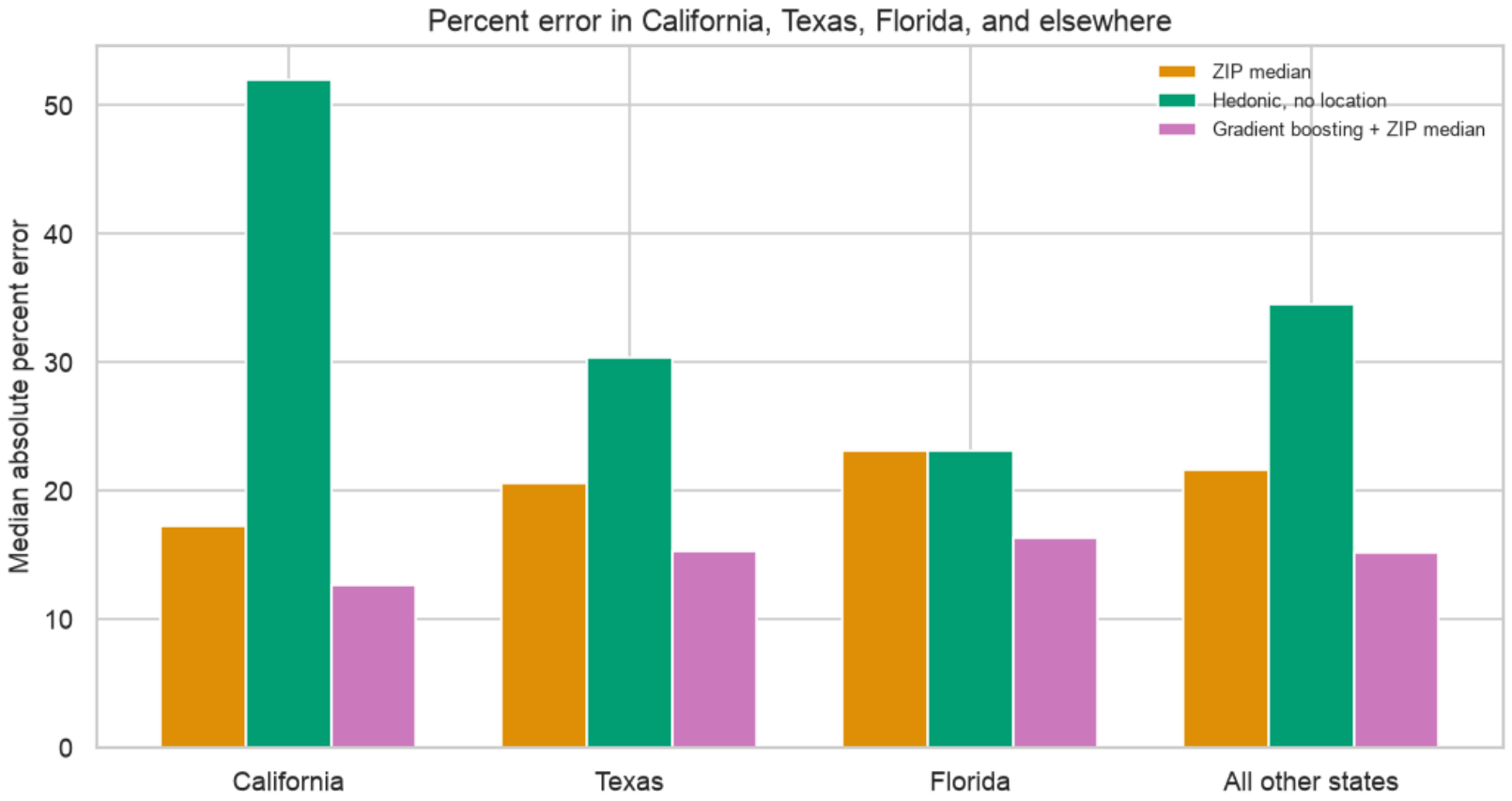
ZIP does most of the work



Thin ZIPs are a different product



California needs its own comps



How to use the number

- Start with the ZIP median. Adjust for living area and bathrooms.
- Show a range. A typical miss is about 15%. About one in three misses by more than 20%.
- Under 20 sales, withhold a firm number.
- Refit before using it outside this window.
- Keep it beside comps. Keep it out of lending.

The answers

- **Accuracy.** About 15% typical miss, and about $99,000 on average, where the ZIP has at least 20 sales.
- **ZIP.** Cuts the typical miss from about 36% to about 15%.
- **Where it changes.** Thin ZIPs are worse. California falls apart without a ZIP, and looks best once the ZIP is included.